

Scientific Computing

Mathematical Foundations for Machine Learning

Course Conspect

Course Structure

The course develops the probability, statistics, linear algebra, calculus, optimization, and numerical computing foundations used throughout machine learning. The detailed programme describes thirteen substantive lecture units. Its original numbering uses “Slot 7” twice; the sequence below normalizes the labels to Lectures 1–13 without removing any listed material.

Block I: Probability and Statistics

Lecture	Topic
1	Probability models, conditioning, and random variables
2	Distributions, expectation, and variance
3	Joint distributions, dependence, and random vectors
4	Samples, concentration, and limit theorems
5	Statistical estimation: method of moments, maximum likelihood, and MAP
6	Sampling distributions, confidence intervals, and bootstrap
7	Entropy, cross-entropy, KL divergence, Fisher information, and the Cramér–Rao bound

Block II: Linear Algebra, Calculus, and Optimization

Lecture	Topic
8	Geometry of data, projections, and least squares
9	Eigenvalues, positive-definite matrices, singular value decomposition, and PCA
10	Multivariable calculus, matrix calculus, and automatic differentiation
11	Optimization for machine learning

Block III: Numerical Computing and Autodiff

Lecture	Topic
12	NumPy, visualization, and numerical linear algebra
13	PyTorch, autograd, and the integrated capstone

The notes are organized as one cumulative conspect: each future lecture will extend the same PDF.

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Chapter 1

Probability Models, Conditioning, and Random Variables

Probability starts with a model: a precise description of what may happen, which statements about the outcome are meaningful, and how probabilities are assigned. Conditioning changes the reference population, while a random variable turns outcomes into numerical values whose distribution can be studied.

1.1 Probability spaces and events

Definition 1.1 (Probability space). A probability space is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Here Ω is the *sample space*, whose elements are outcomes; $\mathcal{F}$ is a collection of subsets of Ω , called events; and $\mathbb{P}$ assigns a probability to every event.

The event collection $\mathcal{F}$ is closed under complements and countable unions:

$$A \in \mathcal{F} \implies A^c \in \mathcal{F}, \quad A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}.$$

It also contains Ω . In a finite model one normally takes $\mathcal{F} = 2^\Omega$, the set of all subsets of Ω .

For events $A, B \subseteq \Omega$:

- $A \cap B$ means that both events occur;
- $A \cup B$ means that at least one occurs;
- $A^c = \Omega \setminus A$ means that A does not occur;
- $A \setminus B = A \cap B^c$ means that A occurs but B does not.

Definition 1.2 (Probability axioms). For events in $\mathcal{F}$, a probability measure satisfies:

$$\begin{aligned} \mathbb{P}(A) &\geq 0, \\ \mathbb{P}(\Omega) &= 1, \\ \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) &= \sum_{i=1}^{\infty} \mathbb{P}(A_i) \quad \text{for pairwise disjoint } A_i. \end{aligned}$$

Several basic rules follow from these axioms:

$$\mathbb{P}(\emptyset) = 0, \quad (1.1)$$

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad (1.2)$$

$$A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B), \quad (1.3)$$

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \quad (1.4)$$

The last identity is the inclusion–exclusion rule. Adding $\mathbb{P}(A)$ and $\mathbb{P}(B)$ counts the overlap twice, so it must be subtracted once.

Example 1.3 (Two fair coin tosses). Toss a fair coin twice and record the ordered result. Then

$$\Omega = \{HH, HT, TH, TT\}, \quad \mathbb{P}(\{\omega\}) = \frac{1}{4} \quad \text{for every } \omega \in \Omega.$$

Let

$$A = \{\text{first toss is } H\} = \{HH, HT\}, \quad B = \{\text{second toss is } H\} = \{HH, TH\}.$$

Then

$$A \cap B = \{HH\}, \quad A \cup B = \{HH, HT, TH\}, \quad A^c = \{TH, TT\}.$$

Inclusion–exclusion gives

$$\mathbb{P}(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}.$$

For a finite sample space with equally likely outcomes,

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

This is a counting shortcut, not an additional axiom: it is valid only after equal likelihood has been justified.

1.2 Conditional probability and total probability

Definition 1.4 (Conditional probability). For $\mathbb{P}(B) > 0$, the probability of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (1.5)$$

Conditioning restricts the reference set to B and renormalizes the remaining probabilities. The event after the vertical bar determines the denominator. In general, $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$ are different quantities.

Rearranging (1.5) gives the multiplication rule

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A). \quad (1.6)$$

Example 1.5 (At least one head). In the two-toss model, let

$$E = \{\text{at least one head}\} = \{HH, HT, TH\}.$$

Given E , the probability that the second toss is heads is

$$\mathbb{P}(B \mid E) = \frac{\mathbb{P}(B \cap E)}{\mathbb{P}(E)} = \frac{2/4}{3/4} = \frac{2}{3}.$$

The unconditional probability is $\mathbb{P}(B) = 1/2$. Conditioning removes the outcome TT , in which B fails.

Theorem 1.6 (Law of total probability). *Let $B_1, \dots, B_m$ be a partition of Ω : the events are pairwise disjoint and their union is Ω . If $\mathbb{P}(B_i) > 0$, then*

$$\mathbb{P}(A) = \sum_{i=1}^m \mathbb{P}(A \mid B_i) \mathbb{P}(B_i). \quad (1.7)$$

Indeed, the events $A \cap B_i$ are disjoint and their union is A . Add their probabilities and then apply the multiplication rule. For a binary partition B, B^c ,

$$\mathbb{P}(A) = \mathbb{P}(A \mid B) \mathbb{P}(B) + \mathbb{P}(A \mid B^c) \mathbb{P}(B^c).$$

1.3 Independence and mutual exclusivity

Definition 1.7 (Independence). Events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B). \quad (1.8)$$

If $\mathbb{P}(B) > 0$, this is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.

Independence means that learning whether one event occurred does not change the probability of the other. In the coin example,

$$\mathbb{P}(A \cap B) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2},$$

so the events “first toss is heads” and “second toss is heads” are independent.

Definition 1.8 (Mutual exclusivity). Events A and B are mutually exclusive, or disjoint, if

$$A \cap B = \emptyset.$$

They cannot occur together.

Proposition 1.9. *Two mutually exclusive events with positive probability are not independent.*

Proof. Mutual exclusivity gives $\mathbb{P}(A \cap B) = 0$, while positive probabilities give $\mathbb{P}(A) \mathbb{P}(B) > 0$. Thus (1.8) cannot hold. $\square$

The distinction is essential:

	Mutual exclusivity	Independence
Can both occur?	No	Usually yes
Defining condition	$A \cap B = \emptyset$	$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B)$
Information meaning	One rules out the other	One does not change the other

1.4 Bayes' theorem and base rates

Theorem 1.10 (Bayes' theorem). *Let $B_1, \dots, B_m$ be a partition of Ω . For $\mathbb{P}(A) > 0$,*

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j) \mathbb{P}(B_j)}{\sum_{i=1}^m \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)}. \quad (1.9)$$

The numerator is the probability of the route B_j followed by A . The denominator adds every mutually exclusive route that can produce A .

Example 1.11 (Rare-defect classifier). Let D mean that an item is defective and $+$ mean that a classifier produces a positive flag. Assume

$$\mathbb{P}(D) = 0.01, \quad \mathbb{P}(+ | D) = 0.99, \quad \mathbb{P}(+ | D^c) = 0.05.$$

The probability of a positive result is

$$\mathbb{P}(+) = 0.99(0.01) + 0.05(0.99) = 0.0594.$$

Therefore

$$\mathbb{P}(D | +) = \frac{0.99(0.01)}{0.0594} = \frac{1}{6} \approx 0.167.$$

The same calculation can be expressed with expected counts in a population of 10,000 items:

Actual state	Positive	Negative	Total
Defect D	99	1	100
No defect D^c	495	9,405	9,900
Total	594	9,406	10,000

Only 99 of the 594 positive items are defective. The sensitivity $\mathbb{P}(+ | D) = 0.99$ is not the posterior $\mathbb{P}(D | +)$. The low base rate and the false positives among the much larger non-defective group dominate the denominator.

For a Bayesian classifier with classes $C_1, \dots, C_K$ and observed features x ,

$$\mathbb{P}(C_k | x) = \frac{p(x | C_k)\mathbb{P}(C_k)}{\sum_{j=1}^K p(x | C_j)\mathbb{P}(C_j)}. \quad (1.10)$$

The factors have distinct roles:

- $\mathbb{P}(C_k)$ is the prior or class base rate;
- $p(x | C_k)$ is the class-conditional likelihood;
- $\mathbb{P}(C_k | x)$ is the posterior after observing x .

The maximum a posteriori rule is

$$\hat{C}_{\text{MAP}}(x) = \arg \max_k p(x | C_k)\mathbb{P}(C_k).$$

1.5 Random variables

Definition 1.12 (Random variable). A real-valued random variable is a function

$$X : \Omega \longrightarrow \mathbb{R}$$

such that $\{\omega : X(\omega) \leq x\}$ is an event for every $x \in \mathbb{R}$. In finite examples this event condition is automatic.

The sampled outcome is random; the mapping X is fixed.

Example 1.13 (Number of heads). For two coin tosses, let X be the number of heads. Then

Outcome ω	HH	HT	TH	TT
$X(\omega)$	2	1	1	0

The support $\{0, 1, 2\}$ is the set of possible values of X ; it is not the original sample space.

A *discrete* random variable takes values in a finite or countable set. An *absolutely continuous* random variable assigns probabilities to intervals through a density. Mixed random variables, with both point masses and a continuous part, also exist.

1.6 PMFs, PDFs, and CDFs

1.6.1 Probability mass functions

Definition 1.14 (PMF). For a discrete random variable X , its probability mass function is

$$p_X(x) = \mathbb{P}(X = x), \quad p_X(x) \geq 0, \quad \sum_x p_X(x) = 1.$$

For the number of heads in two fair tosses,

$$p_X(0) = \frac{1}{4}, \quad p_X(1) = \frac{1}{2}, \quad p_X(2) = \frac{1}{4}.$$

The probability of an event is obtained by summing the masses of its values; for example, $\mathbb{P}(X \geq 1) = p_X(1) + p_X(2) = 3/4$.

1.6.2 Cumulative distribution functions

Definition 1.15 (CDF). For every real-valued random variable X , the cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Every CDF is nondecreasing and right-continuous, with

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow \infty} F_X(x) = 1.$$

It determines interval probabilities through

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a). \quad (1.11)$$

For the number of heads,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{4}, & 0 \leq x < 1, \\ \frac{3}{4}, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

For a discrete variable, the jump at x equals its point mass:

$$p_X(x) = F_X(x) - F_X(x^-),$$

where $F_X(x^-)$ denotes the left limit.

1.6.3 Probability density functions

Definition 1.16 (PDF). An absolutely continuous random variable Y has a probability density function f_Y satisfying

$$f_Y(y) \geq 0, \quad \int_{-\infty}^{\infty} f_Y(y) dy = 1,$$

and

$$\mathbb{P}(a < Y \leq b) = \int_a^b f_Y(y) dy. \quad (1.12)$$

Warning. A density value is not a point probability. For an absolutely continuous random variable, $\mathbb{P}(Y = y) = 0$ for every single point, and $f_Y(y)$ may be larger than one. Probability is the area under the density.

The CDF and density are related by

$$F_Y(y) = \int_{-\infty}^y f_Y(t) dt, \quad F'_Y(y) = f_Y(y)$$

at points where the derivative exists.

Example 1.17 (Uniform distribution on the unit interval). Let

$$f_Y(y) = \begin{cases} 1, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ y, & 0 \leq y \leq 1, \\ 1, & y > 1. \end{cases}$$

Thus

$$\mathbb{P}(0.2 < Y \leq 0.7) = \int_{0.2}^{0.7} 1 dy = F_Y(0.7) - F_Y(0.2) = 0.5.$$

1.7 A compact probability dictionary

Question	Object	Operation
Which outcomes are possible?	Sample space Ω	Specify the model
Did a statement occur?	Event $A \subseteq \Omega$	Use set operations
What changes after learning B ?	$\mathbb{P}(A \mid B)$	Restrict and renormalize
What are all routes to evidence?	Total probability	Sum weighted routes
What caused observed evidence?	Bayes' theorem	Reverse the conditional direction
Probability at a discrete value	PMF $p_X(x)$	Sum masses
Probability over a continuous interval	PDF $f_X(x)$	Integrate area
Probability accumulated to the left	CDF $F_X(x)$	Evaluate or subtract CDF values

1.8 Exercises

Exercise 1.1. For a fair die, let $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$. Find $A \cap B$, $A \cup B$, A^c , their probabilities, and $\mathbb{P}(A \mid B)$. Are A and B independent?

Exercise 1.2. A system uses server type A with probability 0.6 and type B with probability 0.4. Their failure probabilities are 0.02 and 0.05, respectively. Find the total failure probability and the posterior probability that a failed server was type B .

Exercise 1.3. For a fair die, show that $A = \{2, 4, 6\}$ and $B = \{1, 2, 3, 4\}$ are independent. Compare this with A and $C = \{1, 3, 5\}$.

Exercise 1.4. A binary classifier has classes C_0, C_1 , with $\mathbb{P}(C_1) = 0.2$, $\mathbb{P}(s \mid C_1) = 0.6$, and $\mathbb{P}(s \mid C_0) = 0.1$. Compute $\mathbb{P}(C_1 \mid s)$ and select the MAP class.

Exercise 1.5. A discrete variable satisfies

$$p_X(-1) = a, \quad p_X(0) = 2a, \quad p_X(2) = a.$$

Find a , write the CDF, and compute $\mathbb{P}(-1 < X \leq 2)$.

Exercise 1.6. Let

$$f_X(x) = \begin{cases} cx, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find c , derive F_X , and calculate $\mathbb{P}(0.2 < X \leq 0.7)$.

Short answers

1. $A \cap B = \{4, 6\}$, $A \cup B = \{2, 4, 5, 6\}$, and $A^c = \{1, 3, 5\}$. Their probabilities are $1/3$, $2/3$, and $1/2$, respectively; $\mathbb{P}(A \mid B) = 2/3$. Since $\mathbb{P}(A \cap B) = 1/3 \neq \mathbb{P}(A)\mathbb{P}(B) = 1/4$, the events are not independent.
2. $\mathbb{P}(F) = 0.032$ and $\mathbb{P}(B \mid F) = 0.625$.
3. $\mathbb{P}(A \cap B) = 1/3 = \mathbb{P}(A)\mathbb{P}(B)$; A and C are mutually exclusive and not independent.
4. $\mathbb{P}(C_1 \mid s) = 0.6$, so the MAP class is C_1 .
5. $a = 1/4$, and

$$F_X(x) = \begin{cases} 0, & x < -1, \\ 1/4, & -1 \leq x < 0, \\ 3/4, & 0 \leq x < 2, \\ 1, & x \geq 2; \end{cases} \quad \mathbb{P}(-1 < X \leq 2) = \frac{3}{4}.$$

6. $c = 2$, $F_X(x) = x^2$ on $[0, 1]$, and $\mathbb{P}(0.2 < X \leq 0.7) = 0.45$.